



Karachi AQI Hub

AI-Powered Air Quality Forecasting & Health Intelligence

System Status

Champion Model: Gradient Boosting

Test R² Score: 0.9113

Test RMSE: 2.39 µg/m³

Hopsworks Feature Store

Feature Group:

karachi_aqi_features (v1)

Selected Features: 18 Predictors

Sync Mode: Hourly GitHub Actions Cron

Refresh Live Data

Built for 10 Pearls Data Science Internship • Karachi, Pakistan

Karachi Air Quality Index & 72-Hour Forecaster

Karachi, Pakistan • Real-Time Satellite & Atmospheric Telemetry • Updated: Thursday, August 27, 2026 • 09:00 AM UTC

LIVE AQI STATUS: MODERATE



PM10 Coarse Dust

62.4 µg/m³

Temperature

29.1 °C

Relative Humidity

70.0 %

Wind Speed

19.4 km/h

Carbon Monoxide (CO)

193 µg/m³

Surface Pressure

1002 hPa

Health Advisory & Action Plan: Air quality is acceptable; moderate health concern for sensitive individuals.

72-Hour (3-Day) AQI Forecast | Exploratory Data Analysis & Trends | SHAP Explainability & Factor Insights | Scenario Simulator Sandbox | Model Leaderboard & MLOps Registry

72-Hour (3-Day) Recursive Air Quality Forecast

Sequential multi-step forecast simulating meteorological transitions and diurnal smog dynamics for Karachi.

Day 1 (Thursday, Aug 27)

27.9 µg/m³

Status: Moderate

Peak: 27.9 µg/m³ (Thu 07:00 PM)

Day 2 (Friday, Aug 28)

27.9 µg/m³

Status: Moderate

Peak: 27.9 µg/m³ (Fri 09:00 AM)

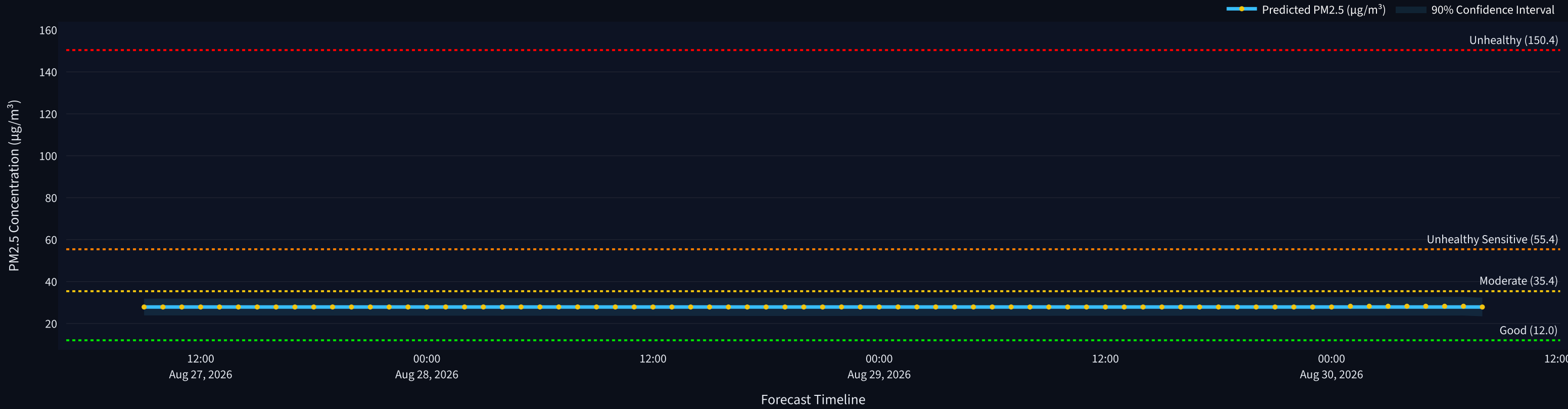
Day 3 (Saturday, Aug 29)

28.0 µg/m³

Status: Moderate

Peak: 28.2 µg/m³ (Sun 01:00 AM)

Next 72-Hour PM2.5 Forecast Curve with Confidence Interval



View Hourly Forecast Data Table

	timestamp	forecast_hour_display	predicted_pm25	lower_bound_90	upper_bound_90	aqi_category	health_advisory
0	2026-08-27 09:00:00+00:00	Thu 09:00 AM	27.87	23.87	31.87	Moderate	Air quality is acceptable; moderate health concern for sensitive individuals.
1	2026-08-27 10:00:00+00:00	Thu 10:00 AM	27.87	23.85	31.89	Moderate	Air quality is acceptable; moderate health concern for sensitive individuals.
2	2026-08-27 11:00:00+00:00	Thu 11:00 AM	27.87	23.83	31.91	Moderate	Air quality is acceptable; moderate health concern for sensitive individuals.
3	2026-08-27 12:00:00+00:00	Thu 12:00 PM	27.87	23.81	31.93	Moderate	Air quality is acceptable; moderate health concern for sensitive individuals.
4	2026-08-27 13:00:00+00:00	Thu 01:00 PM	27.87	23.8	31.94	Moderate	Air quality is acceptable; moderate health concern for sensitive individuals.
5	2026-08-27 14:00:00+00:00	Thu 02:00 PM	27.87	23.79	31.95	Moderate	Air quality is acceptable; moderate health concern for sensitive individuals.
6	2026-08-27 15:00:00+00:00	Thu 03:00 PM	27.87	23.78	31.96	Moderate	Air quality is acceptable; moderate health concern for sensitive individuals.
7	2026-08-27 16:00:00+00:00	Thu 04:00 PM	27.87	23.76	31.98	Moderate	Air quality is acceptable; moderate health concern for sensitive individuals.
8	2026-08-27 17:00:00+00:00	Thu 05:00 PM	27.87	23.75	31.99	Moderate	Air quality is acceptable; moderate health concern for sensitive individuals.
9	2026-08-27 18:00:00+00:00	Thu 06:00 PM	27.87	23.74	32	Moderate	Air quality is acceptable; moderate health concern for sensitive individuals.